Cobalt Instrument Studio

Engineering Report on Simulation and Verification

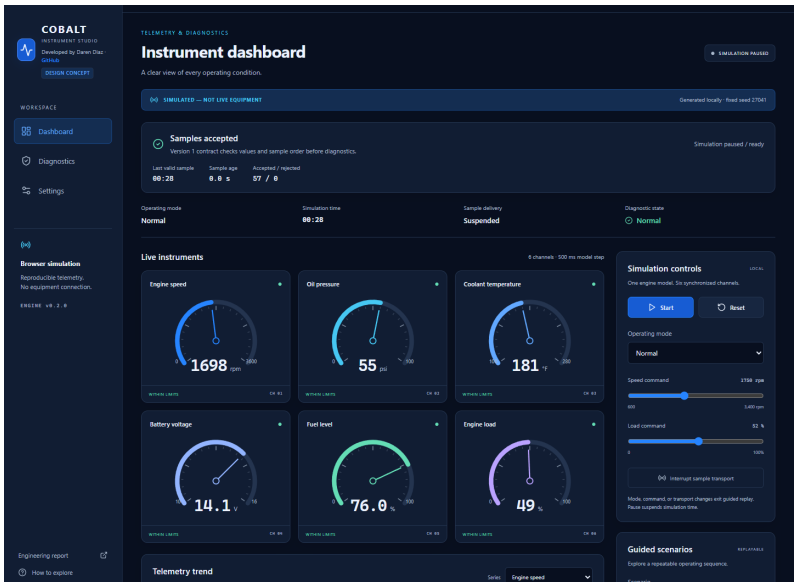
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Version 0.2.0 | Fall 2026

Purpose and verified checkpoints

Cobalt Instrument Studio is an original static React and TypeScript simulator. One seeded model drives accepted readings, trends, and diagnostics. Versioned telemetry validation protects six instruments across five guided scenarios. Version 0.2.0 adds Gauges, Engine, Trends, Faults, and ACK device pages with a graphite bezel, metallic rings, cobalt blue accents, and large white readouts.

The earlier core revision was merged through PR 1 at main 453ea4, with 92 unit and 12 Linux Chromium browser tests and verified production sites. The current local version 0.2.0 functionality and privacy correction have separate evidence. Publication through reviewed PRs, main CI, updated hosting, and a new annotated tag is approved after final validation. The initial v0.1.0 Release is preserved.



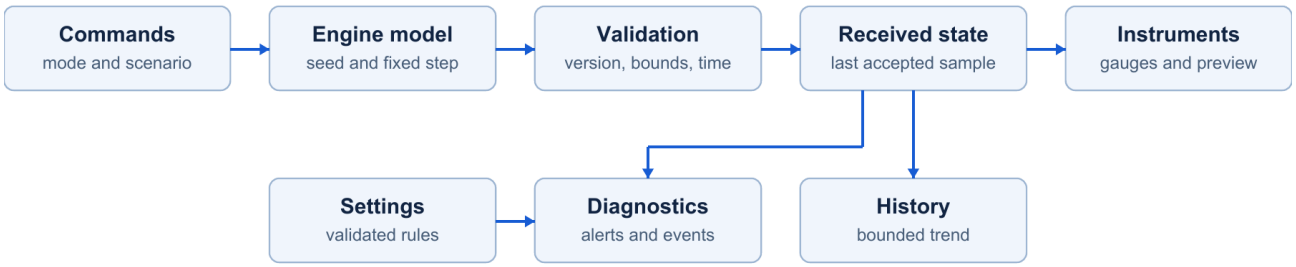
Actual local dashboard capture. Instrument readings and the device preview consume simulated accepted data.

Simulation scope

SIMULATED - NOT LIVE EQUIPMENT. Every value is generated locally. There is no backend, authentication, equipment connection, hardware control, certification, or production guarantee. Demonstration relationships and plausibility ranges are not calibrated equipment measurements or safe operating limits.

Architecture and state ownership

One typed reducer owns the simulation session. Commands advance a fixed-step seeded engine; simulated transport then supplies an unknown sample envelope to validation. Only accepted canonical telemetry reaches shared readings, metric rules, history, gauges, and the device preview. Rejected deliveries leave that accepted state intact while time and freshness age continue advancing.



The acceptance boundary separates generated process values from accepted display and diagnostic data.

Component	Local capability	Engineering value
Engine and scenarios	One seed, six coupled channels, fixed 0.5 s step, five replays	Reproducible process and delivery fault paths
Telemetry contract	Version, fields, numeric bounds, and timestamp ordering	Rejected values cannot contaminate accepted state
Diagnostics	Thresholds, hysteresis, debounce, timeout, acknowledgment	Explained condition onset, escalation, and recovery
History and events	240 accepted samples and 100 event records	Deliberate retention and counted/deduplicated evidence
Device workspace	Six signal bindings, display properties, five preview pages	Presentation can change independently of process rules
Settings and export	Validated local preferences and simulated CSV	Inspectable configuration and portable session evidence

Pure modules under src/engine define model, validation, rules, and display layout. Browser LocalStorage holds validated application preferences; telemetry, events, and device layout stay in session memory. Operation requires no server API, device runtime, or remote data source.

Versioned telemetry contract

The flat version 1 envelope requires contractVersion, time, rpm, oilPressure, coolantTemp, batteryVoltage, fuelLevel, and engineLoad as own properties. Version must be numeric 1. Channel/time values must be finite numbers; no string coercion is performed. Additional fields are ignored and are never copied into accepted telemetry.

Field	Canonical unit	Inclusive plausibility bounds
time	Simulation seconds	At least 0; no future time; strictly newer than last accepted sample
rpm	rpm	0 to 6000
oilPressure	psi	0 to 150
coolantTemp	Degrees F	-40 to 320
batteryVoltage	V	0 to 32
fuelLevel	Percent	0 to 100
engineLoad	Percent	0 to 100

Rejected delivery and freshness

Malformed envelopes, unsupported versions, missing fields, wrong types, nonfinite/out-of-bounds values, negative/future time, duplicate timestamps, and out-of-order timestamps each produce an explicit rejection reason. A rejected attempt does not replace readings, update accepted time, append history, or count toward metric debounce. Age is measured from the last accepted timestamp.

Each rejection code is logged once per continuous rejection episode; repeats increase a counter without repeated messages. An accepted delivery records one recovery and resets the episode. A sustained invalid episode can therefore produce both sample-quality evidence and the normal communication timeout. Events remain capped at 100.

Distinct bounds and clocks

These are demonstration plausibility limits, broader than the generator and display ranges. A narrowed preview scale preserves the numeric reading and clamps its needle. The 500 ms timer is a scheduling target, not a real-time guarantee. Pause freezes simulation time and age; reset/replay establishes a fresh ordering baseline. See docs/TELEMETRY_CONTRACT.md for the complete acceptance policy.

Reproducible process behavior

Normal operating ramp

Expected: increase idle speed and load smoothly while pressure follows and temperature responds slowly. Observed seed 27041 at 35 simulated seconds: 1749.8 rpm, 51.8 percent load, 57.1 psi oil pressure, and 182.7 degrees F coolant. Default rules remained normal. Replay resets accepted data, ordering, history, and diagnostics while retaining valid application preferences.

Coolant overheat

Expected: cooling degradation starts at 6 s and crosses warning and critical. The retained seed/default policy reaches warning at 14.0 s (223.5 degrees F) and critical at 18.5 s (236.8 degrees F). Thresholds are 220/235 degrees F, with 4 degrees F hysteresis and two accepted samples of debounce. Escalation renews acknowledgment; reset creates a normal session.

Oil pressure loss and recovery

Expected: lower generated pressure from 5 s, cross 24 psi warning and 16 psi critical, then restore generation at 24 s. Observed warning was 11.5 s/22.199 psi; critical 15.5 s/14.726 psi; recovery through warning 26.0 s/23.410 psi; normal 27.5 s/30.110 psi. The 3 psi recovery margin and two accepted samples govern clearing. Its 42 s replay accepted 85 samples with no rejection; browser checks exercise acknowledgment and renewed acknowledgment after escalation.

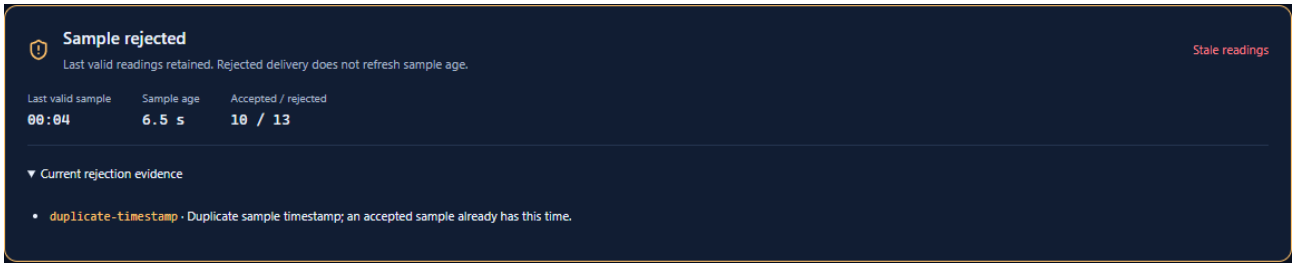
Rule behavior	Engineering meaning
Valid process fault	Low pressure and high coolant are valid samples; diagnostics must detect them rather than reject their delivery.
Acknowledgment	Records review, leaves an active condition intact, and resets after escalation or recovery.
Hysteresis/debounce	Separate recovery limits and consecutive accepted samples limit threshold chatter.
Seeded replay	Fixed simulation steps reproduce identical accepted readings and transition sequences.

All times describe the simulator. Thermal and pressure fault kinetics are accelerated explanatory assumptions, not real machinery response or field calibration.

Sample delivery faults and recovery

Scenario	Expected accepted-state behavior
Communication loss	No delivery from 5 s through 16.5 s. Retain the 4.5 s sample; default timeout begins at 6.5 s. Valid delivery resumes at 17 s.
Invalid sample delivery	Reject nonfinite, missing/wrong-type, duplicate, and out-of-order attempts from 5 s through 16.5 s. Retain accepted values/history; timeout and recovery follow accepted time.

Both scenarios distinguish the internal generated model from accepted readings. A missing or rejected sample cannot hide age by replacing receivedAt. The trend leaves a gap over missing accepted measurements. A valid sample at 17 s restores freshness and communication normal state; invalid-delivery recovery also closes the rejection episode.



Actual local browser capture of invalid sample delivery: visible rejection evidence and retained accepted readings.

Expected versus observed evidence

Observed invalid replay: 24 rejected attempts; first reason events at 5, 8, 9.5, 11, and 14 s; retained 4.5 s readings; timeout at 6.5 s; one delivery recovery and communication clear at 17 s. At 25 s, accepted count/history were 27. Unit/browser checks verify isolation, age, counts, acknowledgment, and recovery. Guided browser clocks do not establish real network or hardware reliability.

Device pages and usable state

The workspace combines five device pages, a six-signal browser, and a selected-slot property editor. A graphite bezel, metallic gauge rings, cobalt blue accents, and large white readouts make the preview resemble an instrument panel. Accepted readings, bounded history, and central diagnostics supply every page. Display bindings, labels, and ranges affect presentation without changing engine commands or alarm thresholds.



Actual local Engine page capture. Blue cockpit styling presents accepted simulated readings inside the device preview.

Property or behavior	Implemented limit
Pages and softkeys	Gauges, Engine, Trends, Faults, and ACK; page controls and device softkeys share navigation state.
Standby lifecycle	Initial/reset readouts show zero in either unit system; Start or guided Run reveals accepted values, and Pause holds them.
Trends and acknowledgment	Trends use accepted bounded history. ACK invokes central alert acknowledgment and preserves active conditions and escalation rules.
Binding and display range	One existing signal per slot; finite ordered range within that signal display domain; canonical psi/degrees F editor units.
Label and apply	Trimmed 1 to 32 characters, no control characters; Apply is atomic and Reset restores the selected slot.
Out-of-scale value	Numeric reading remains visible; preview labels above/below scale and clamps the needle.
Persistence	Six fixed slots and version 1 session layout; no save/import/export or real display runtime.

Standby zeros are display placeholders and do not enter telemetry, diagnostics, or accepted history. Current local checks covered 30 device-page/theme/width combinations and the existing 18 application accessibility combinations, including standby/start/pause/reset, softkeys, metric display, stale readings, central ACK and typography, with no JavaScript errors. A formal assistive-technology audit remains future work.

Local validation and measured evidence

Starting baseline: 32 unit and 7 browser tests passed. Core hardening recorded 58 unit and 12 browser tests; the earlier DevOps PR passed 92 unit and 12 Linux Chromium browser tests. Current local version 0.2.0 evidence records 100 passing unit tests and 17 passing browser tests. Each checkpoint has separate evidence; local counts do not establish new remote execution.

Requirement	Method and actual outcome	Status
Contract isolation	Invalid fields, finite/bounds checks, version and time ordering; accepted readings/history remain uncontaminated.	Pass
Five seeded replays	Retained scenarios plus oil loss and invalid delivery; deterministic transition/recovery assertions.	Pass
Diagnostic rules	Acknowledgment, escalation, hysteresis, clearing, rejected-sample debounce and timeout recovery.	Pass
Bounded evidence	240 accepted samples; 100 events; repeated rejection reasons counted without floods.	Pass
Device properties	Atomic valid apply, invalid property rejection, binding/range/label checks, model/rule independence.	Pass
Browser workflows	Standby/start/pause/reset, five scenarios, device pages/softkeys, central ACK, preferences, keyboard/focus, reduced motion and mobile fit.	Pass
Two static bases	Root and project-base production build paths; nonempty matching PDF/DOCX downloads.	Pass
Report layout	Final PDF and DOCX rendered and every page inspected with approved local tools.	Pass
Privacy and dates	Developer/repository credit, blank report metadata, Fall 2026 edition; disabled Git capture and sanitized retained test reports.	Pass

Measured observations

Root production JavaScript totaled 659,335 bytes (198,235 bytes gzip); CSS totaled 54,619 bytes. Browser tests took 37.8 s; summed test-file durations were 3.85 s. One real-time desktop smoke observed 3 simulated seconds after 3498 ms including interaction overhead; this is one workflow measurement, not scheduling accuracy.

Exact counts and measurements are retained in local validation evidence. Public checks at main 453ea4 verified both origins and report hashes. Temporary test artifacts exposed unnecessary Git contact metadata; approved deletion was verified and a local capture/sanitization correction was added. No hardware, Safari/Firefox, physical cellular, long-soak, frame-rate, or uptime result is claimed.

Production gaps and engineering tradeoffs

The hardening revision makes invalid data, reproducible faults, and display configuration reviewable before equipment integration. A real deployment needs separate connectivity, safety, security, calibration, and operational evidence.

Area	Implemented demonstration	Future production requirement
Input and time	Versioned generated data; plausibility and simulation-time ordering	Validated device identity, units, clock origin, sequence counters, framing, integrity, and schema migration
Connectivity	Simulated missing/rejected delivery	Gateway adapter, protocol testing and controlled network impairment; CAN/J1939 remains hypothetical
Safety/security	Display/simulator only; no equipment or credentials	Hazard analysis, separated safety/control, authentication, permissions, secret lifecycle, and independent review
Diagnostics	Thresholds, hysteresis, debounce, acknowledgment and delivery evidence	Calibrated alert policy, nuisance-alarm analysis, validated operator response and routing
Device workspace	Session-only five-page preview, six gauge slots and display properties	Validated portable layout, supported target runtime, arbitrary widgets and accessible layout editing
Records and reliability	Bounded memory, simulated CSV, local automated checks	Durable audited records, monitoring, cross-browser/device tests, hardware tests, soak, rollback and service objectives

Deliberate tradeoffs

A static frontend avoids server, credential, and equipment dependencies. A central reducer exposes replay/state ownership. Strict acceptance preserves trustworthy displayed data at the cost of stale readings when delivery is invalid. Fixed steps favor repeatability over real-time fidelity. Session-only buffers and display layout keep retention explicit; their caps do not prove a total heap bound.

The application cannot command a physical machine, prove a sensor fault, certify an alarm policy, or replace operational monitoring. The Design concept label identifies an independent engineering demonstration.

Verified workflow and publication boundaries

Published core and local follow up

PR 1 published telemetry validation, five guided scenarios, device configuration, accessibility changes, and inspectable CI. Version 0.2.0 adds five device pages with blue cockpit styling and corrected test-evidence handling. The central engine, accepted-sample boundary, three application views, simulation disclaimer, display-only properties, preferences, and two static hosting bases are preserved.

Prioritized future engineering

P0: real gateway adapter; calibrated device schema/identity; hardware in the loop; connectivity security/isolation; Safari/Firefox and physical mobile coverage; formal accessibility audit.

P1: jitter/intermittent/simultaneous faults; durable history; validated alarm routing; accountable configuration audit; independent device streams; versioned layout import/export; authorized physical configuration transactions.

P2: additional widgets/layout editing; target display runtime/export; long-session performance profiling; release provenance/rollback; production observability; calibrated simulation models. The roadmap records benefit and complexity for 19 specific future items and separately marks implemented work.

Publication boundary

Version 0.2.0 publication is approved after final QA and passing CI. The earlier main 453ea4 passed PR/main CI and both providers were verified. Main protection requires PRs, strict CI from GitHub Actions, administrator enforcement and resolved conversations; reviewer approvals are zero. Two affected temporary test artifacts were removed after approval. New UI/privacy publication and release results are recorded only after actual execution.

The annotated v0.1.0 tag and its initial two report assets are unchanged. Version 0.2.0 aligns package, changelog, notes, and report metadata. The tag workflow verifies that agreement and reruns CI before publishing one Release with engineering PDF/DOCX, a tested static archive, manifest, and SHA-256 checksums. Its actual new-version execution and downloads remain to be verified.

Source: https://github.com/CloudWizardBoricua/cobalt-instrument-studio. Primary: https://cobalt-instrument-studio.netlify.app/. Backup: https://cloudwizardboricua.github.io/cobalt-instrument-studio/. README links actual PR 1, PR CI 38341235636, main CI 38341542078, Pages 38341678181, and the initial Release. The workflow evidence files separately record artifact inspection, production behavior, protection, and cleanup.